AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 8 · STUDENT LESSON

Features: Useful Clues

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 7: agree on the target label before selecting features.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Choose recorded properties that preserve distinctions needed for a stated task. Try the recall before opening notes.

Your learning target

I can choose recorded properties that preserve distinctions needed for a stated task.

I can show why identical **feature record**s can hide examples with different **target label**s.

I can reject an answer copied into the **input**s and test a proposed **feature** with a new case.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

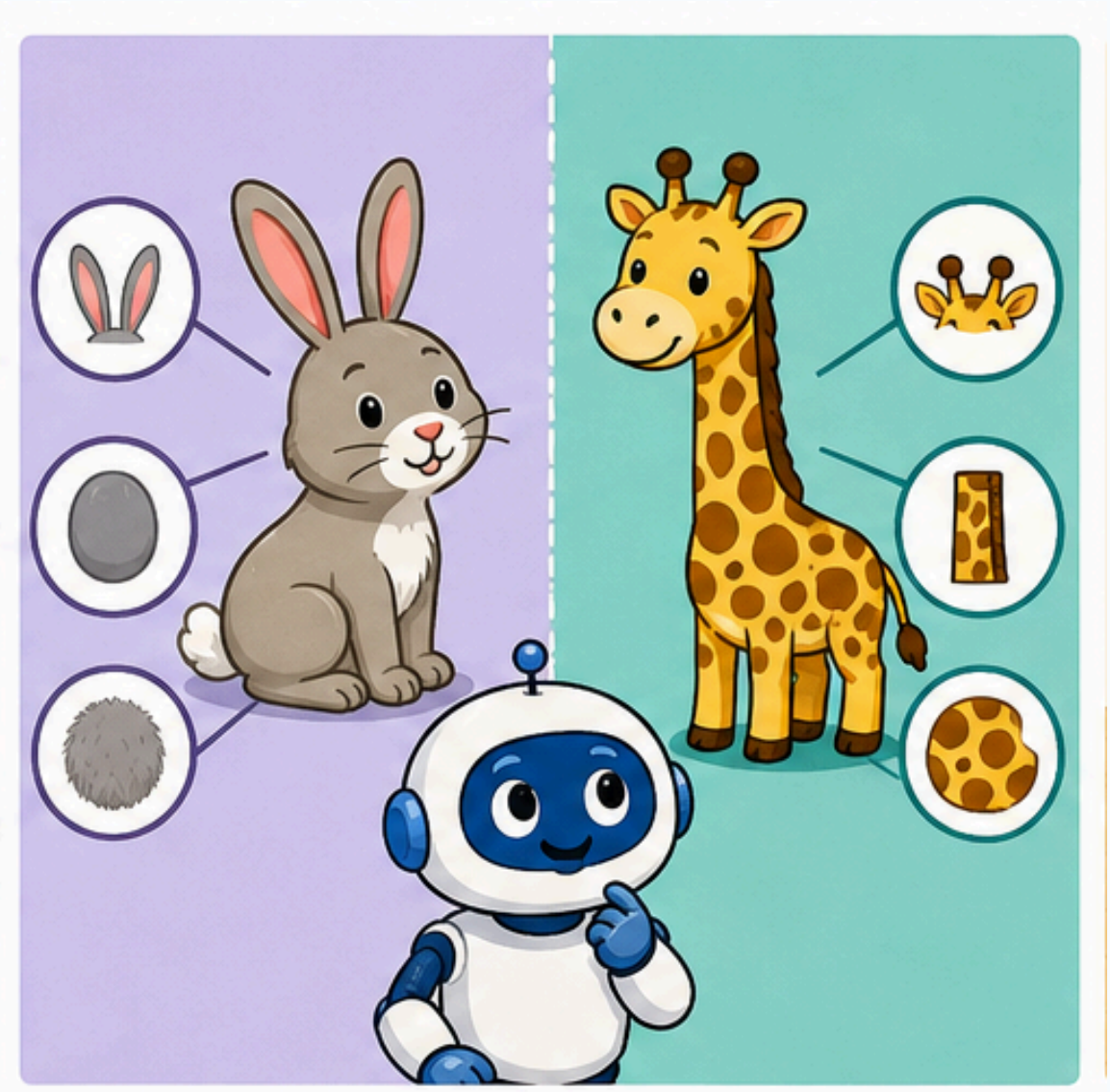
Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

After trying, check the **relevant** lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Some observable clues help a decision more than others.



Your target

LEARNING TARGET A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Learner name:

Readiness for the connected explanation

Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

This week's end goal

By the end, I can choose task-**relevant feature**s and avoid a clue created after the decision.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Features: Useful Clues case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or **distraction**.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Features: Useful Clues. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



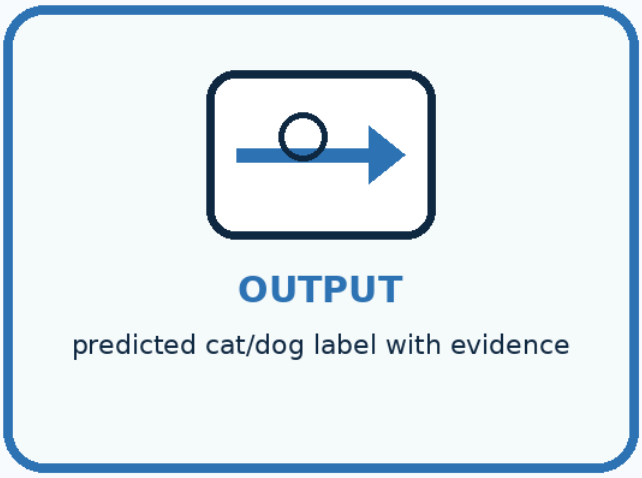
INPUT

measured candidate clues



PROCESS

compare useful patterns and ignore distractions



INPUT measured candidate clues

PROCESS compare useful patterns and ignore **distraction**s

OUTPUT predicted cat/dog label with **evidence**

Key idea

CORE IDEA Some observable clues help a decision more than others.

How the mechanism works

UNDER THE HOOD A **feature** is a represented property used by a decision **process**. Predictive usefulness depends on the task and data.

Spurious **feature**s correlate in a sample without reflecting the intended concept and often fail when conditions change.

Draw a fresh example with all three stages:

A closer explanation

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

The same cards under two feature choices

ORIGINAL CARD	CHOOSE THE INPUTS	RESULTING FEATURE RECORD
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

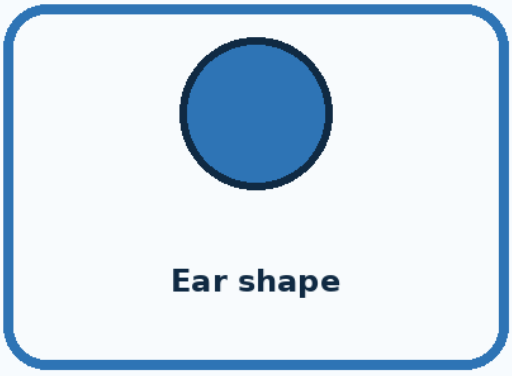
Compare the cases

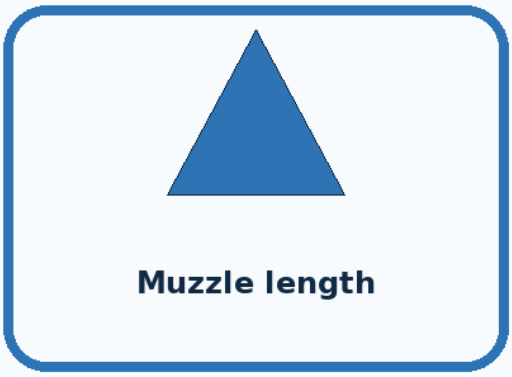
> Compare the supplied cases for Features: Useful Clues. Identify the detail that changes the conclusion.

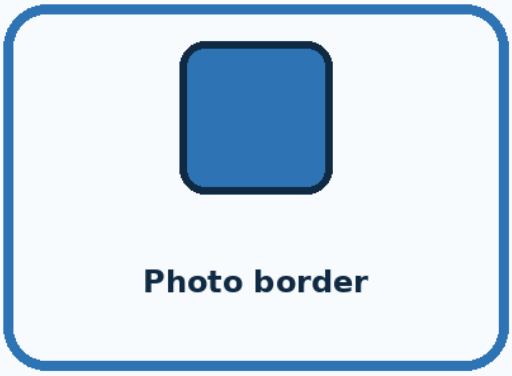
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Ear shape Clue: may differ across many examples. Meaning: potentially useful.

The case to investigate

CASE 2 — Muzzle length Clue: observable animal trait. Meaning: potentially useful.

The case to investigate

CASE 3 — Photo border Clue: added by the photographer. Meaning: **distraction**.

The case to investigate

CASE 4 — Grass background Clue: accidental setting pattern. Meaning: **spurious clue**.

Keep this boundary clear

BOUNDARY A clue can be observable yet still be ir**relevant**, unfair, or fragile.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Change Q's width to 6 without changing its height.

Predicted result

Prediction: Q fits the shelf.

Why the result follows

Reason: Both dimensions now meet their including equality limits; stripe pattern does not change the rule.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

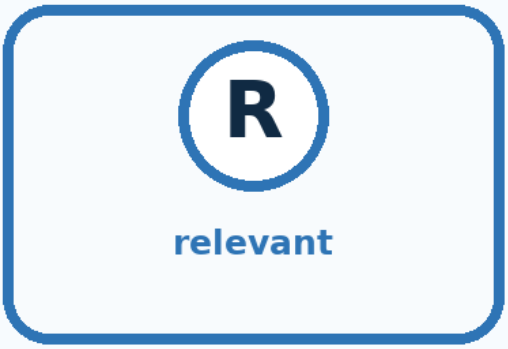
> Use the vocabulary of Features: Useful Clues to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
feature	represented clue used in a decision	
relevant	connected to the task goal	

Word	Meaning	My example
distraction	available clue that should not drive the decision	
spurious clue	accidental pattern that may fail later	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR Use every available clue because **more information** is always better.

Use one vocabulary word and one piece of **evidence** to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

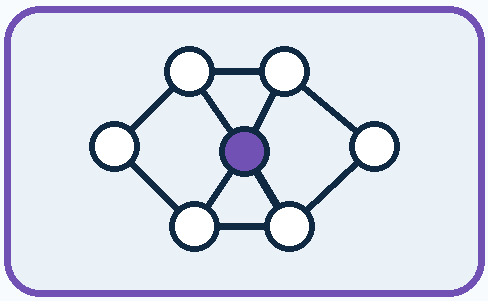
Follow a worked example

> Follow the worked trace for Features: Useful Clues, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

Step 1

STEP 1 — OBSERVE measured candidate clues

Step 2

STEP 2 — TRACE compare useful patterns and ignore **distraction**s

Step 3

STEP 3 — CONCLUDE predicted cat/dog label with **evidence**

Model answer

MODEL REASONING Useful **feature**s describe the animal and remain available across settings. Borders and backgrounds describe the photo setup.

Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

A claim to avoid

DO NOT OVERCLAIM A clue can be observable yet still be ir**relevant**, unfair, or fragile.

What new **evidence** could change this answer?

A closer explanation

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

Worked explanation · Features: Useful Clues

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Worked explanation · Features: Useful Clues

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the shortcut.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features: Useful Clues

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

The facts we will use

Supplied facts

Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

Supplied facts

A green acceptance sticker is attached only after the shelf check.

AI ACADEMY / AI-1/8

Dimensions determine shelf acceptance

	Width cm	Height cm	Outcome
P stripes	6	4	fits
Q stripes	7	4	width fails
R dots	6	5	height fails
S dots	6	4	fits

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/8

Features: Useful Clues

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Width check

P,Q,R,S have widths 6,7,6,6.

2. Height check

Their heights are 4,4,5,4.

3. Apply both limits

P and S fit; Q fails width; R fails height.

4. Reject a shortcut

The later sticker cannot be an input available before checking.

CHANGE ONE THING

Change Q's width to 6 without changing its height.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Width check	P,Q,R,S have widths 6,7,6,6.
2. Height check	Their heights are 4,4,5,4.
3. Apply both limits	P and S fit; Q fails width; R fails height.
4. Reject a shortcut	The later sticker cannot be an input available before checking.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce.

What this example does not establish

Keep the stated limit

These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

Keep the stated limit

Read the labelled picture: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

Keep the stated limit

Change one thing: Change Q's width to 6 without changing its height. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Features: Useful Clues claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Ear shape	may differ across many examples	potentially useful
Muzzle length	observable animal trait	potentially useful
Photo border	added by the photographer	distraction
Grass background	accidental setting pattern	spurious clue

Repair two tempting claims

Challenge claim A

CLAIM A Use every available clue because **more information** is always better.

Repair:

Challenge claim B

CLAIM B A **feature** that worked once will always work.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Before adopting a **feature** set, state the task, name the **input**s with units, and ask whether two different answers can share the same **input** record. Construct such a pair if you can.

Then check that each **input** is available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

Check the claim against the facts

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly.

That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the **feature**s again.

More columns are not automatically better; each one needs a reason to be collected and used.

Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead

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STUDENT LESSON · 8 OF 18

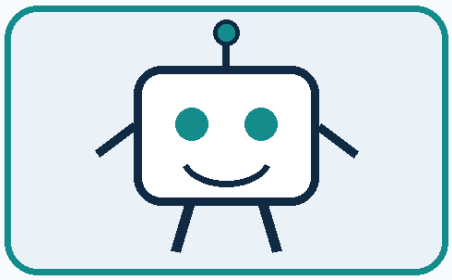
Practise together

Practise Features: Useful Clues with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Some observable clues help a decision more than others.

Use a fresh example to explain **feature**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A **feature** that worked once will always work.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an **input** to predict the label before that inspection.

The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended **prediction**. This is **leakage**: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker.

Keep labels available for teaching and checking **prediction**s, but do not feed the sticker back in as an **input** for this task.

A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label

Practise with less help: Prove that one measurement is insufficient

Use workbook W2 for the connected example "**Feature**s and useful distinctions". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: "A stronger model can always recover an omitted measurement". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

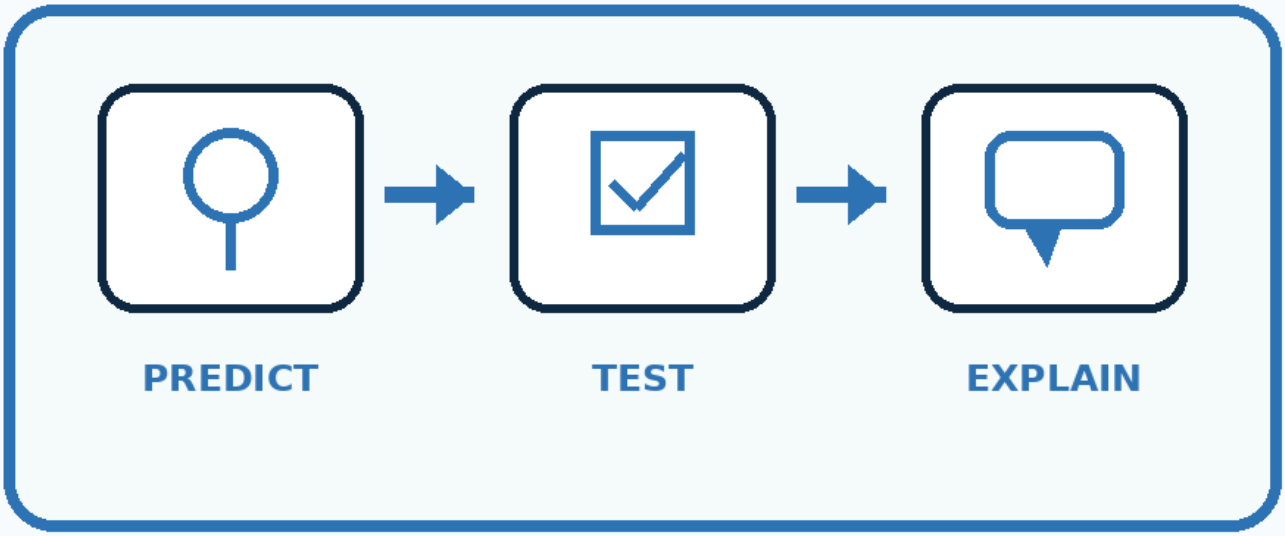
Run the investigation

> Investigate Features: Useful Clues. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Clue Keep-or-Reject

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Choose three clues for cat/dog sorting; reject **distraction**s.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

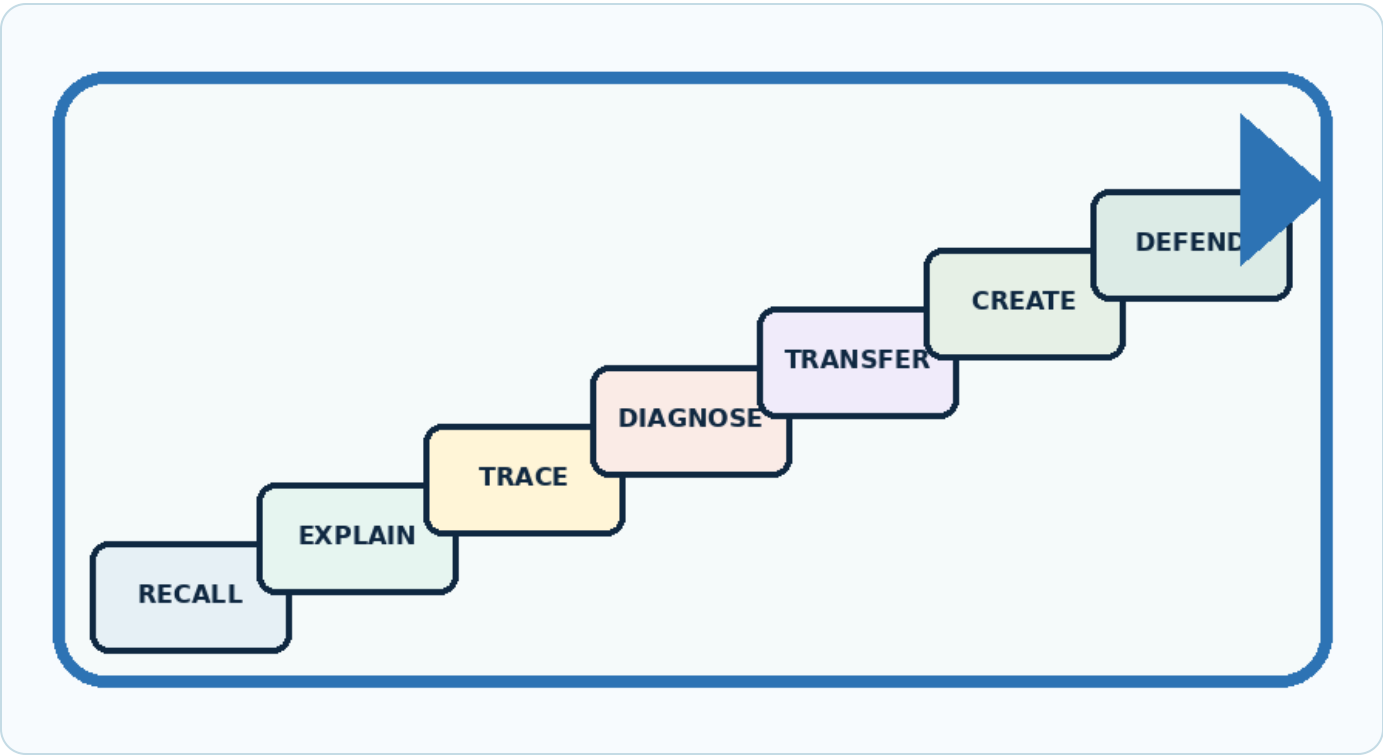
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Features: Useful Clues. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **feature**.

EXPLAIN Some observable clues help a decision more than others.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Use every available clue because **more information** is always better.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

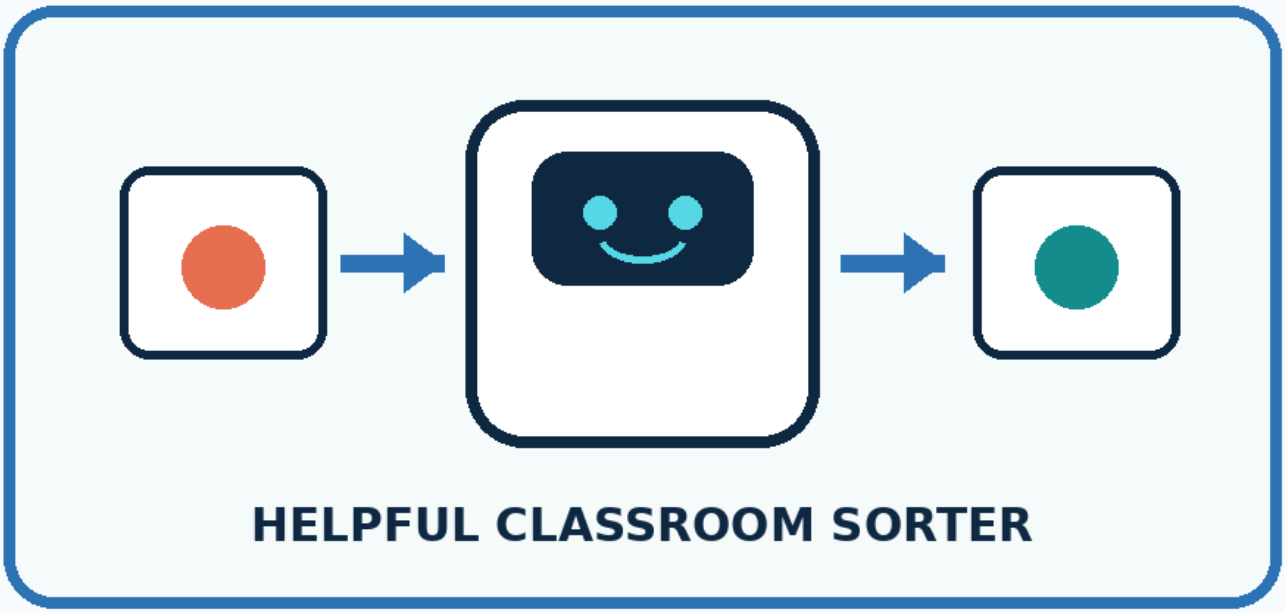
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

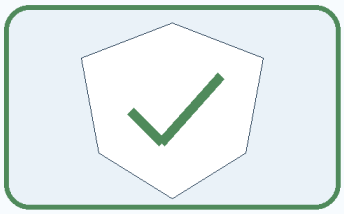
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A clue can be observable yet still be ir**relevant**, unfair, or fragile.

My project decision:

Evidence for your project

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement.

STUDENT LESSON · 12 OF 18

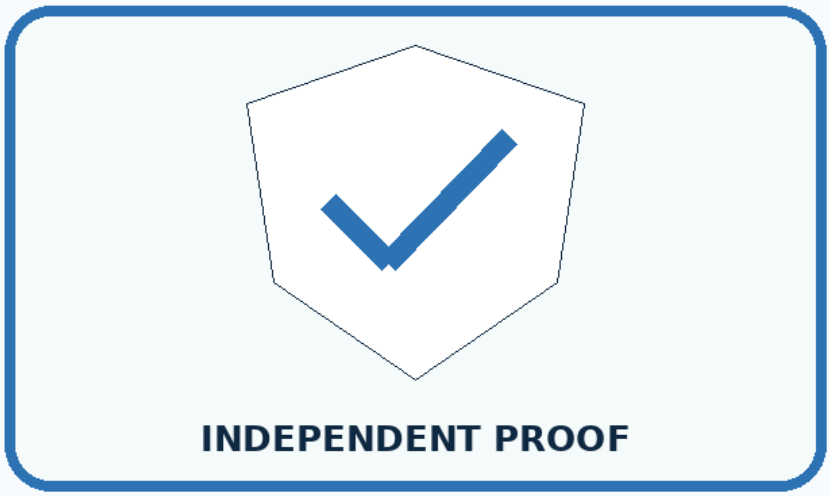
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Select three **feature**s for new cat/dog cards, reject two **distraction**s,

and predict one case where a chosen **feature** would fail.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Useful **feature**s describe the animal and remain available across settings. Borders and backgrounds describe the photo setup.

Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Algorithms as Recipes.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Features: Useful Clues case for supported practice. Repeated examples

> do not become fresh assessments.

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

The question

Choose the **relevant feature** and explain why size is insufficient.

Worked explanation

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Choose a clue that answers this task

Evidence or stage	Value or result
Target	Smooth versus toothed edge

Evidence or stage	Value or result
Useful clue	Edge shape
Weak clue	Leaf size alone

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the **relevant feature** and explain why size is insufficient.

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful **feature** and a distracting **feature**. Test your choice on a blue circle.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one **relevant** detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Features: Useful Clues, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

By the end, I can	How I will show it
Choose recorded properties that preserve distinctions needed for a stated task	Explain the deciding step, show a trace or test, and state one limit.
Show why identical feature records can hide examples with different target labels	Explain the deciding step, show a trace or test, and state one limit.
Reject an answer copied into the inputs and test a proposed feature with a new case	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

Finish and return

> Explain Features: Useful Clues without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose recorded properties that preserve distinctions needed for a stated task

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The later sticker cannot be an **input** available before checking.

1. Apply both limits

2. Reject a shortcut

3. Height check

4. Width check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject a shortcut The later sticker cannot be an **input** available before checking. Comparing P with Q isolates width; comparing P with R isolates height.

Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**. A later acceptance sticker leaks the very answer the system should produce.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change Q's width to 6 without changing its height. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Q fits the shelf.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Q fits the shelf. Both

> dimensions now meet their including equality limits; stripe pattern does not change the rule.

Repair a missing building block

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

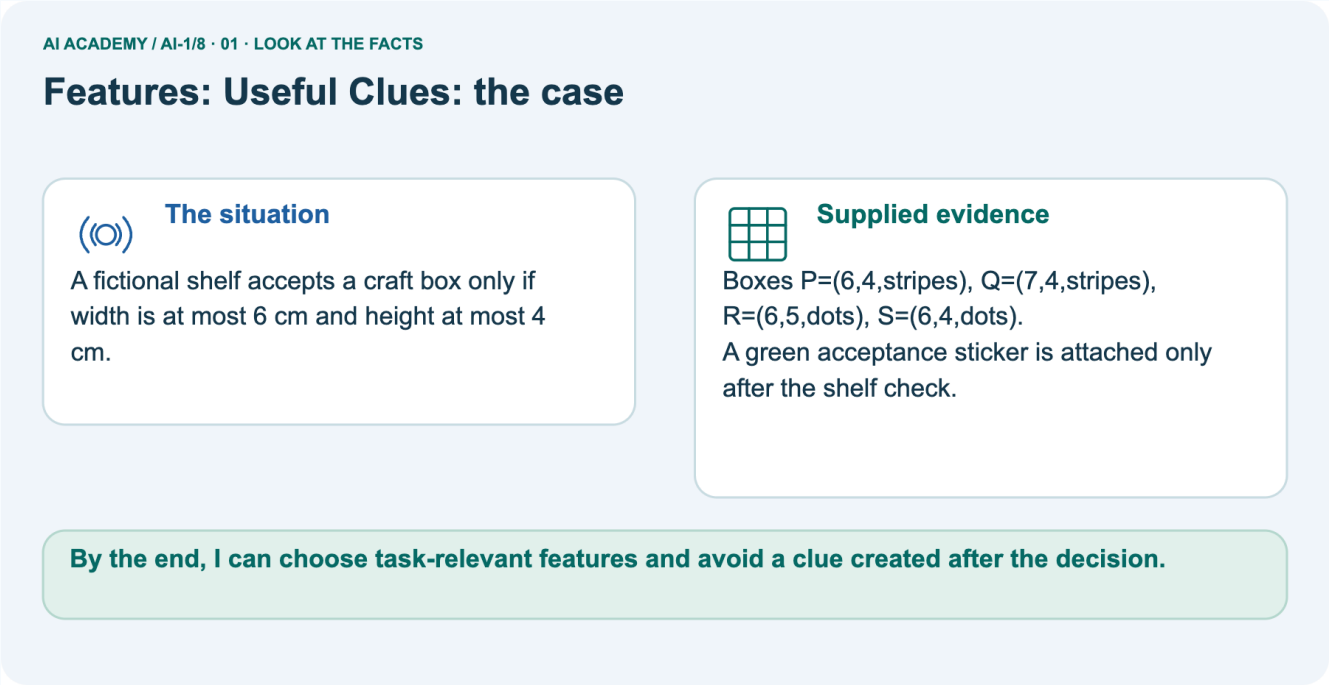
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Features: Useful Clues graphic using its labels, deciding evidence and relevant limitation.

01. Features: Useful Clues: the case



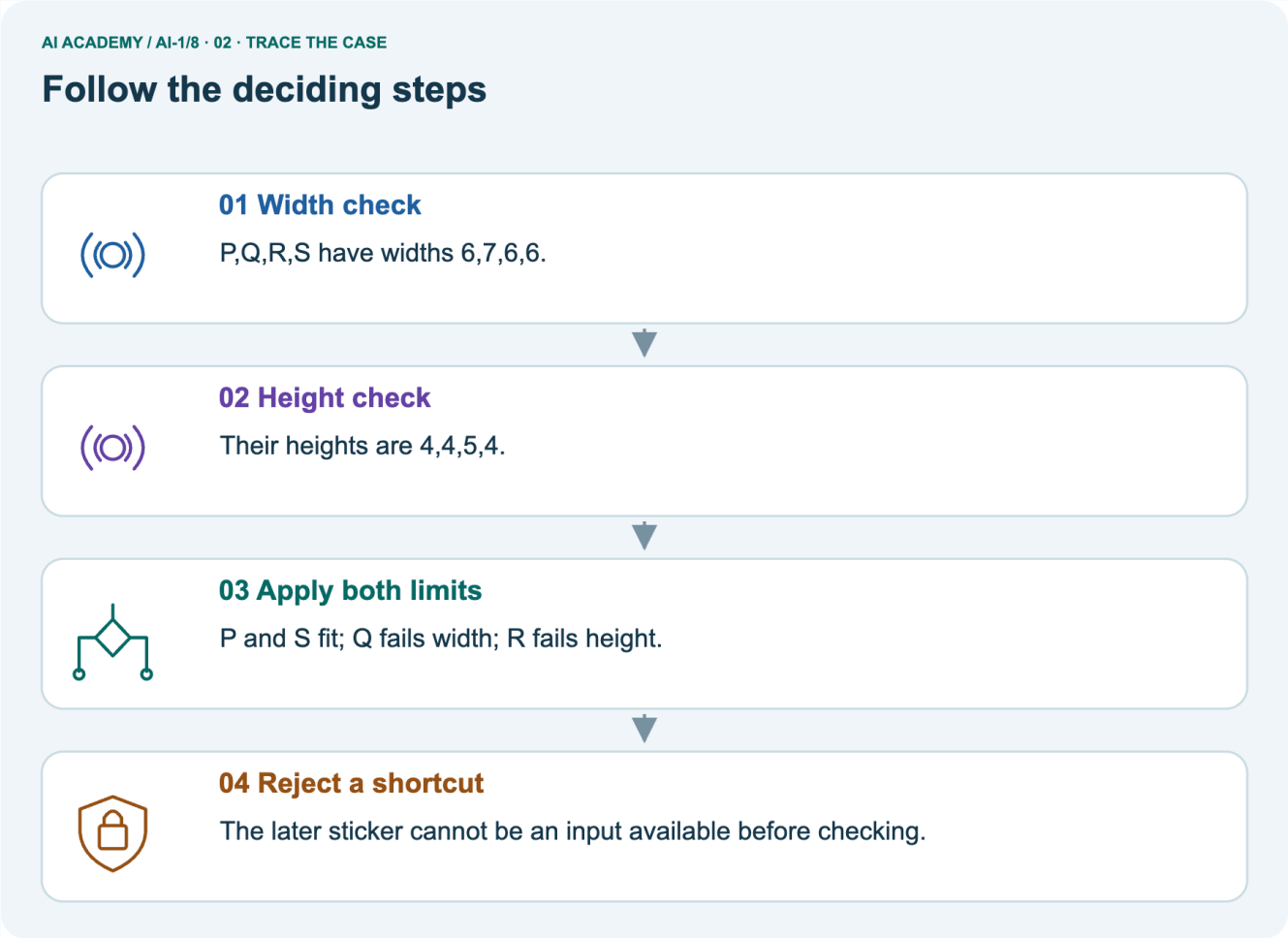
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm. Supplied facts: Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

A green acceptance sticker is attached only after the shelf check.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Width check: P,Q,R,S have widths 6,7,6,6. → Height check: Their heights are 4,4,5,4. → Apply both limits: P and S fit; Q fails width; R fails height. → Reject a shortcut: The later sticker cannot be an **input** available before checking.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Dimensions determine shelf acceptance



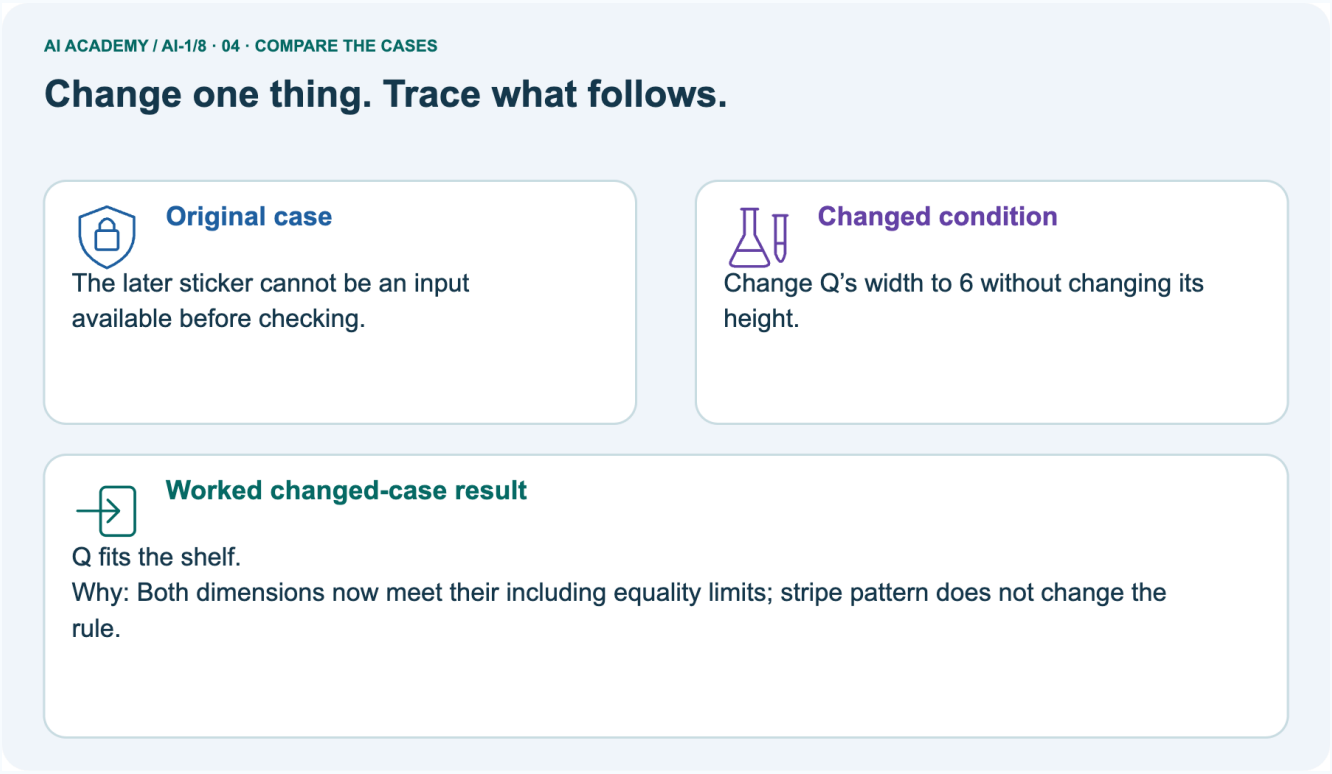
Follow the labelled picture with the matching explanation. Open the vector diagram

Dimensions determine shelf acceptance. P stripes: Width cm = 6, Height cm = 4, Outcome = fits. Q stripes: Width cm = 7, Height cm = 4, Outcome = width fails. R dots: Width cm = 6, Height cm = 5, Outcome = height fails.

S dots: Width cm = 6, Height cm = 4, Outcome = fits.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



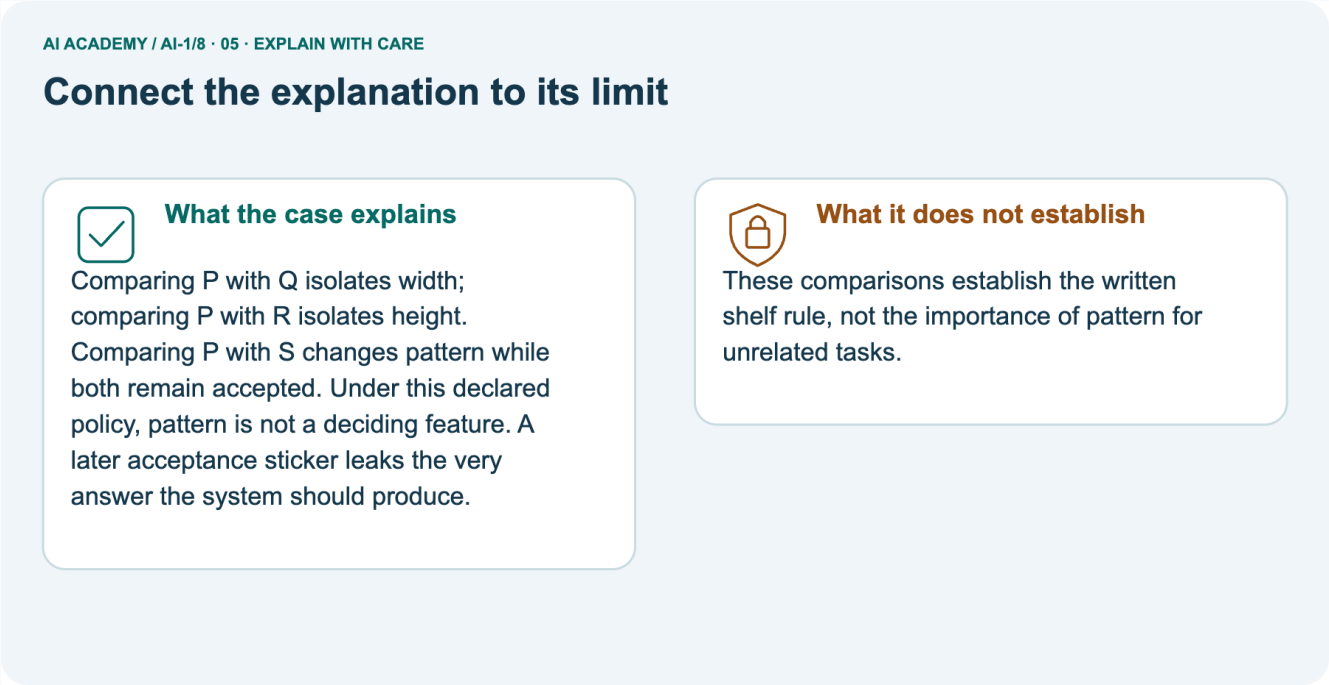
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The later sticker cannot be an **input** available before checking. Changed condition: Change Q's width to 6 without changing its height. Expected result: Q fits the shelf.

Reason: Both dimensions now meet their including equality limits; stripe pattern does not change the rule.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



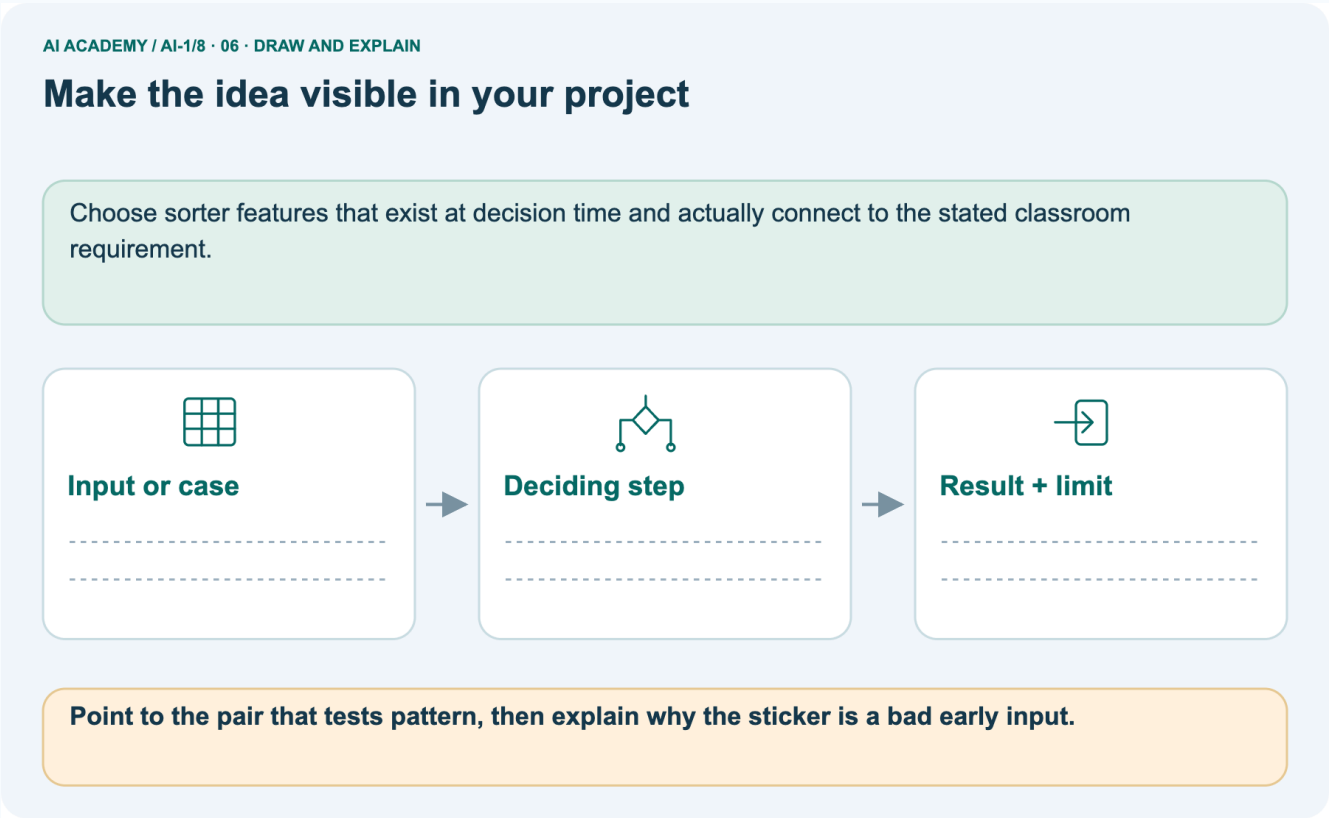
Follow the labelled picture with the matching explanation. Open the vector diagram

Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce. Limit: These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement. Draw your **input**, deciding step, result and limit.

Supported question: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.